Olympiad Test Paper — Mathematics (Class 7)

Chapter 2: Arithmetic Expressions

Subject	Mathematics (NCERT Class 7)
Chapter	2 — Arithmetic Expressions
Total Questions	60 (Multiple Choice, single correct)
Maximum Marks	240
Suggested Time	90 minutes

Marking scheme: **+4** for each correct answer · **−1** for each wrong answer · **0** if unattempted.

Instructions: Each question has four options (A), (B), (C), (D); exactly one is correct. Always do brackets first, then × and ÷, then + and −; never evaluate strictly left-to-right. Watch the classic traps: a minus sign in front of a bracket flips every inner sign, distributing means multiplying every term inside, and only addition/multiplication may be reordered freely — subtraction and division may not. Many items can be compared or simplified without full computation. Do not write on this paper — mark answers on your answer sheet.

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1. What is the value of 30 + 5 × 4?

(A) 50 (B) 140
(C) 35 (D) 20

2. Which of these is an arithmetic expression with the value 12?

(A) 2 × 5 (B) 24 ÷ 2
(C) 20 − 6 (D) 6 + 7

3. Evaluate: 100 − (15 + 56).

(A) 141 (B) 71
(C) 29 (D) 39

4. The terms of the expression 23 − 2 × 4 + 16 are:

(A) 23, 2, 4, 16 (B) 23, −2, ×4, +16
(C) 23 − 2, 4 + 16 (D) 23, −(2 × 4), 16

5. Without computing, which is greater: 1023 + 125 or 1022 + 128?

(A) 1022 + 128 (B) 1023 + 125
(C) They are equal (D) Cannot be decided

6. Evaluate: 6 × 3 − 4 × 8 × 5.

(A) 142 (B) −142
(C) 718 (D) −18

7. Which expression has a value DIFFERENT from the other three?

(A) 3×4 (B) $15 - 3$
(C) $24 \div 3$ (D) $10 + 2$

8. Evaluate: $30 - 5 \times 4 + 2$.

(A) 102 (B) 100
(C) 22 (D) 12

9. Removing the brackets, $500 - (250 - 100)$ equals:

(A) $500 - 250 - 100$ (B) $500 - 250 + 100$
(C) $500 + 250 - 100$ (D) $500 + 250 + 100$

10. By the distributive property, $2 \times (5 + 3)$ equals:

(A) $2 \times 5 + 2 \times 3$ (B) $2 \times 5 + 3$
(C) $2 + 5 \times 2 + 3$ (D) $2 \times 5 \times 2 \times 3$

11. Which is greater: 23×99 or $23 \times 100 - 23$?

(A) 23×99 (B) $23 \times 100 - 23$
(C) Cannot be decided (D) They are equal

12. Irfan buys a ₹15 biscuit packet and ₹56 of dal and pays with ₹100. The expression for his change is:

(A) $100 - 15 + 56$ (B) $(100 - 15) + 56$
(C) $100 - (15 + 56)$ (D) $100 + 15 - 56$

13. Use the distributive property to compute 7×103.

(A) 721 (B) 700
(C) 730 (D) 731

14. Evaluate: $50 - (18 - 6 + 4)$.

(A) 22 (B) 30
(C) 34 (D) 42

15. How many of these expressions have the value 20: $(10 + 10)$, (4×5), $(25 - 5)$, $(60 \div 2)$?

(A) 1 (B) 3
(C) 2 (D) 4

16. Madhu's drone goes 6 m up then 4 m down. Which pair of expressions both give the final height?

(A) $6 - 4$ and $4 - 6$ (B) $6 \times (-4)$ and $(-4) \times 6$
(C) $6 + 4$ and $4 + 6$ (D) $6 + (-4)$ and $(-4) + 6$

17. Evaluate: $6 + 2 \times (15 - 3 \times 4) \div 2$.

(A) 12 (B) 9

(C) 15 (D) 6

18. Manasa added a long list of numbers and got 11749, then realised she had left out 9055. The corrected total is:

(A) 20804 (B) 2694

(C) 11749 (D) 20794

19. Which regrouping makes $25 \times 17 \times 4$ easiest to do mentally, and what is the product?

(A) $(25 \times 17) \times 4 = 425$ (B) $(17 \times 4) \times 25 = 680$

(C) $25 \times (17 \times 4) = 425$ (D) $(25 \times 4) \times 17 = 1700$

20. The value of $(30 + 5) \times 4$ is:

(A) 50 (B) 35

(C) 140 (D) 120

21. Which statement is TRUE?

(A) $30 + (5 \times 4) = (30 + 5) \times 4$

(B) Brackets never change the value of an expression

(C) $30 + (5 \times 4)$ and $(30 + 5) \times 4$ have different values

(D) $30 + 5 \times 4 = 140$

22. Evaluate: $200 - (40 + 3)$.

(A) 157 (B) 163

(C) 237 (D) 143

23. Which is the correct list of terms of the expression $83 - 14 + 5$?

(A) 83, 14, 5 (B) 83, −14, 5

(C) 83, −14, −5 (D) 83 − 14, 5

24. Subtraction is NOT commutative. Which equality is therefore FALSE?

(A) $6 + 4 = 4 + 6$ (B) $(-4) + 6 = 6 + (-4)$

(C) $3 \times 5 = 5 \times 3$ (D) $6 - 4 = 4 - 6$

25. Evaluate: $(-7) + 10 + (-11)$.

(A) −8 (B) 8

(C) −14 (D) 28

26. To compute 102×45 using the distributive property, the best split is:

(A) $100 \times 45 + 2$ (B) $100 \times 45 + 2 \times 45$

(C) 102×45 cannot be split (D) $102 \times 45 = 100 \times 45$ only

27. Which expression equals $100 - 15 - 56$?

(A) 100 – (15 – 56) (B) 100 + (15 + 56)
(C) (100 – 15) + 56 (D) 100 – (15 + 56)

28. Evaluate: 8 + 12 ÷ 4 – 2.

(A) 3 (B) 7
(C) 9 (D) 18

29. A common mistake gives 2 × (5 + 3) = 13. What error was made?

(A) Multiplied 2 by 5 only, then added 3 (B) Added before multiplying
(C) Multiplied 2 by 3 only, then added 5 (D) Subtracted instead of adding

30. Which is greater: 113 – 25 or 112 – 24?

(A) 113 – 25 (B) 112 – 24
(C) They are equal (D) Cannot be decided

31. Evaluate: 12 – 2 × 3 + 4 × 5.

(A) 24 (B) 26
(C) 56 (D) 14

32. Which expression correctly represents "subtract the sum of 8 and 5 from 40"?

(A) 40 – 8 + 5 (B) 8 + 5 – 40
(C) (40 – 8) + 5 (D) 40 – (8 + 5)

33. By the distributive property, 6 × 98 equals:

(A) 6 × 100 – 6 × 2 (B) 6 × 100 – 2
(C) 6 × 100 + 6 × 2 (D) 6 × 90 + 8

34. Evaluate: 500 – 250 + 100.

(A) 150 (B) 350
(C) 650 (D) 850

35. Which expression does NOT equal 24?

(A) 4 × (2 + 4) (B) 30 – 6
(C) 4 × 2 + 4 (D) 48 ÷ 2

36. In the expression 9 – 4 × 2, after writing it as a sum of terms, the terms are:

(A) 9 and −4 and 2 (B) 5 and 2
(C) 9 and 8 (D) 9 and −8

37. Without full computation, which is greater: 47 + 38 or 49 + 35?

(A) 47 + 38 (B) 49 + 35
(C) They are equal (D) Cannot be decided

38. Evaluate: 3 × (8 – 2 × 3) + 5.

(A) 23 (B) 41
(C) 11 (D) 5

39. Which is equal to $7 \times (10 - 3)$?

(A) $7 \times 10 - 3$ (B) $7 \times 10 - 7 \times 3$
(C) $7 \times 10 + 7 \times 3$ (D) $70 - 3$

40. Evaluate: $60 \div (2 + 3) \times 2$.

(A) 6 (B) 18
(C) 12 (D) 24

41. Reema writes $18 - 5 + 3$ and gets 10; her brother gets 16. Who is correct and why?

(A) The brother — $18 - 5 = 13$, then $13 + 3 = 16$, so 16 is correct
(B) The brother — you always go right to left
(C) Reema — subtract the sum $5 + 3 = 8$ first, giving 10
(D) Both are correct depending on the day

42. The value of $25 \times 4 + 25 \times 6$ (using distributivity) is:

(A) 100 (B) 150
(C) 250 (D) 1000

43. Which expression has the GREATEST value?

(A) $2 + 3 \times 4$ (B) $2 \times 3 \times 4$
(C) $2 \times 3 + 4$ (D) $(2 + 3) \times 4$

44. Evaluate: $100 - 3 \times (12 - 2 \times 5)$.

(A) 194 (B) 6
(C) 90 (D) 94

45. The expression $15 - (7 - 2)$ is equal to:

(A) $15 - 7 + 2$ (B) $15 - 7 - 2$
(C) $15 + 7 - 2$ (D) $15 + 7 + 2$

46. A shopkeeper sells 8 pens at ₹12 each and 5 notebooks at ₹20 each. The total bill (in ₹) is the value of:

(A) $8 + 12 \times 5 + 20$ (B) $8 \times 12 + 5 \times 20$
(C) $(8 + 5) \times (12 + 20)$ (D) $8 \times 5 + 12 \times 20$

47. Which is greater without computing: 999×8 or $1000 \times 8 - 8$?

(A) 999×8 (B) $1000 \times 8 - 8$
(C) They are equal (D) Cannot be decided

48. Evaluate: $4 \times 5 - 2 \times (6 + 1)$.

(A) 126 (B) 26
(C) 34 (D) 6

49. Which removal of brackets is CORRECT?

(A) $80 - (20 - 5) = 80 - 20 - 5$ (B) $80 - (20 - 5) = 80 - 20 + 5$
(C) $80 - (20 - 5) = 80 + 20 - 5$ (D) $80 - (20 - 5) = 80 + 20 + 5$

50. The expression 5×25 represents Mallika spending ₹25 each school day from Monday to Friday. Its value is:

(A) ₹125 (B) ₹30
(C) ₹100 (D) ₹150

51. Evaluate: $2 \times (3 + 4 \times (2 + 1))$.

(A) 14 (B) 42
(C) 30 (D) 18

52. Which expression equals 45×99?

(A) $45 \times 100 - 1$ (B) $45 \times 100 + 45$
(C) $45 \times 90 + 9$ (D) $45 \times 100 - 45$

53. Evaluate: $36 \div 4 \div 3$.

(A) 3 (B) 27
(C) 12 (D) 9

54. Which pair of expressions is EQUAL?

(A) $(12 - 4) - 2$ and $12 - (4 - 2)$ (B) $12 - 4 - 2$ and $12 - (4 + 2)$
(C) $12 \div 4 \div 2$ and $12 \div (4 \div 2)$ (D) $12 - 4 + 2$ and $12 - (4 + 2)$

55. Evaluate: $7 + 3 \times 4 - 5 \times 2$.

(A) 30 (B) 35
(C) 14 (D) 9

56. Sunita has the expression $64 - (24 + 16)$ and rewrites it without brackets. Which is correct?

(A) $64 - 24 + 16 = 56$ (B) $64 + 24 - 16 = 72$
(C) $64 - 24 - 16 = 24$ (D) $64 + 24 + 16 = 104$

57. Evaluate: $10 + 2 \times 5 - 3 \times (4 - 2)$.

(A) 14 (B) 54
(C) 6 (D) 18

58. To rearrange $8 + 17 + 2 + 13$ for easy mental addition, the grouping that pairs numbers into round tens is:

(A) $(8 + 17) + (2 + 13)$ (B) $(8 + 2) + (17 + 13)$
(C) $(8 + 13) + (17 + 2)$
(D) cannot be regrouped — addition order is fixed

59. Evaluate: $(15 - 3 \times 4) + (6 \div 2 + 1) \times 2$.

(A) 5 (B) 3
(C) 11 (D) 19

60. Which statement about the value of $36 - 12 \div 4 + 2 \times 5$ is TRUE?

(A) It equals 36, because you work left to right(B) It equals 40
(C) It equals 33 (D) It equals 43, doing ÷ and × before − and +

End of question paper. Maximum marks: 240. Marking: +4 correct, −1 wrong, 0 unattempted.

Solutions — Mathematics (Class 7), Chapter 2: Arithmetic Expressions

Marking: **+4** correct, **−1** wrong, **0** unattempted. Maximum marks **240**.

Answer Key

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	A	11	D	21	C	31	B	41	A	51	C
2	B	12	C	22	A	32	D	42	C	52	D
3	C	13	A	23	B	33	A	43	B	53	A
4	D	14	C	24	D	34	B	44	D	54	B
5	A	15	B	25	A	35	C	45	A	55	D
6	B	16	D	26	B	36	D	46	B	56	C
7	C	17	B	27	D	37	A	47	C	57	A
8	D	18	A	28	C	38	C	48	D	58	B
9	B	19	D	29	A	39	B	49	B	59	C
10	A	20	C	30	C	40	D	50	A	60	D

Answer-key distribution: A = 15, B = 15, C = 15, D = 15.

Worked Solutions

1. (A) ×/÷ before +/−: 5 × 4 = 20 first, then 30 + 20 = 50. Evaluating left-to-right (30 + 5 = 35, × 4) gives the trap 140.

2. (B) 24 ÷ 2 = 12. The others: 2 × 5 = 10, 20 − 6 = 14, 6 + 7 = 13. Only 24 ÷ 2 hits 12.

3. (C) Do the bracket first: 15 + 56 = 71, then 100 − 71 = 29. (Dropping the bracket as 100 − 15 + 56 = 141 is the classic minus-bracket trap.)

4. (D) Terms are the parts separated by + after turning subtraction into adding the inverse, and a product stays whole. So 23 − 2 × 4 + 16 has terms 23, −(2 × 4), 16. (Splitting 2 × 4 into separate terms is the trap.)

5. (A) Compare the differences instead of adding: 1022 + 128 starts 1 lower (1022 vs 1023) but gains 3 more (128 vs 125), a net +2, so 1022 + 128 is the larger sum. Check: 1023 + 125 = 1148 and 1022 + 128 = 1150. ✓

6. (B) Two terms: 6 × 3 = 18 and 4 × 8 × 5 = 160. The expression is 18 − 160 = −142. (Forgetting the sign gives 142.)

7. (C) 3 × 4 = 12, 15 − 3 = 12, 10 + 2 = 12, but 24 ÷ 3 = 8. The odd one out is 24 ÷ 3.

8. (D) Multiply first: 5 × 4 = 20. Then 30 − 20 + 2 = 10 + 2 = 12. (Left-to-right with 30 − 5 = 25 then × 4 gives traps like 102/100.)

9. (B) A minus before the bracket flips every inner sign: 500 − (250 − 100) = 500 − 250 + 100. The −100 inside becomes +100.

10. (A) Distribute the 2 over both terms: 2 × (5 + 3) = 2 × 5 + 2 × 3. Multiplying only the first term (2 × 5 + 3) is the common slip.

11. (D) 23 × 100 − 23 = 23 × (100 − 1) = 23 × 99 by the distributive property, so the two are exactly equal — no computation of the product needed.

12. (C) Total the purchases first, then subtract from 100: change = 100 − (15 + 56). The bracket forces the sum before subtraction.

13. (A) 7 × 103 = 7 × 100 + 7 × 3 = 700 + 21 = 721. (Forgetting the 7 × 3 term gives 700/730.)

14. (C) Bracket first: 18 − 6 + 4 = 16. Then 50 − 16 = 34. (50 − 18 − 6 + 4 = 30 mishandles the inner signs.)

15. (B) Values: 10 + 10 = 20 ✓, 4 × 5 = 20 ✓, 25 − 5 = 20 ✓, 60 ÷ 2 = 30 ✗. So 3 of the 4 equal 20.

16. (D) Up 6 then down 4 is 6 + (−4) = 2 m, and by commutativity (−4) + 6 = 2 m as well — same height. 6 − 4 and 4 − 6 are NOT equal (2 vs −2), so that pair fails.

17. (B) Innermost first: 3 × 4 = 12, so 15 − 12 = 3. Then 2 × 3 ÷ 2 = 3 (×/÷ left to right). Finally 6 + 3 = 9.

18. (A) By the associative/commutative properties she need not restart — just add the missing number to the running total: 11749 + 9055 = 20804.

19. (D) Regroup so 25 pairs with 4: (25 × 4) × 17 = 100 × 17 = 1700 — easy mental maths. (25 × 17) gives 425, not 1700, so option A is wrong; 17 × 4 = 68 makes B/C arithmetic wrong too.

20. (C) The bracket forces 30 + 5 = 35 first, then × 4 = 140. (Without the bracket, 30 + 5 × 4 would be 50 — option A is that trap.)

21. (C) 30 + (5 × 4) = 50 but (30 + 5) × 4 = 140, so they differ — brackets DO change the value here. (A) and (D) are false; (B) is false because brackets can change a value.

22. (A) Bracket first: 40 + 3 = 43, then 200 − 43 = 157.

23. (B) Convert the subtraction to adding the inverse: $83 - 14 + 5 = 83 + (-14) + 5$, so the terms are 83, −14, 5. The +5 stays positive.

24. (D) Subtraction is not commutative: $6 - 4 = 2$ but $4 - 6 = -2$, so $6 - 4 = 4 - 6$ is false. The other three (addition and multiplication) are genuinely commutative.

25. (A) Group: $(-7 + 10) + (-11) = 3 + (-11) = -8$. (Adding magnitudes as $7 + 10 + 11 = 28$ ignores the signs.)

26. (B) $102 \times 45 = (100 + 2) \times 45 = 100 \times 45 + 2 \times 45$. Distributing over BOTH parts. "$100 \times 45 + 2$" forgets to multiply the 2 by 45.

27. (D) Working backwards, $100 - 15 - 56 = 100 - (15 + 56)$ (a minus in front of the bracket distributes the subtraction to both terms). $100 - (15 - 56)$ would be $100 - 15 + 56$.

28. (C) ÷ before +/−: $12 \div 4 = 3$. Then $8 + 3 - 2 = 9$.

29. (A) Getting 13 means doing $2 \times 5 = 10$ then adding the bare 3: $10 + 3 = 13$ — multiplying only the first term inside the bracket. The correct distribution is $2 \times 5 + 2 \times 3 = 16$.

30. (C) Compare the differences: the right side is 1 smaller on top (112 vs 113) AND subtracts 1 less (24 vs 25), a net change of 0. So $113 - 25 = 112 - 24 = 88$. They are equal.

31. (B) Terms: 12, $-(2 \times 3) = -6$, $+(4 \times 5) = +20$. Sum = $12 - 6 + 20 = 26$.

32. (D) "The sum of 8 and 5" is $(8 + 5)$; "subtract it from 40" is $40 - (8 + 5)$. Without the bracket, $40 - 8 + 5 = 37$ only subtracts the 8.

33. (A) $6 \times 98 = 6 \times (100 - 2) = 6 \times 100 - 6 \times 2 = 600 - 12 = 588$. Both terms get the 6.

34. (B) Same precedence, left to right: $500 - 250 = 250$, then $250 + 100 = 350$.

35. (C) $4 \times (2 + 4) = 24$, $30 - 6 = 24$, $48 \div 2 = 24$, but $4 \times 2 + 4 = 8 + 4 = 12$. So $4 \times 2 + 4$ is the one that is NOT 24.

36. (D) Rewrite as a sum of terms: $9 - 4 \times 2 = 9 + (-(4 \times 2)) = 9 + (-8)$. The terms are 9 and −8. (4×2 is one whole term.)

37. (A) Differences: the left's first number is 2 smaller (47 vs 49) but its second is 3 larger (38 vs 35) — net +1 — so $47 + 38$ is greater. ($47 + 38 = 85$, $49 + 35 = 84$.)

38. (C) Inner term: $2 \times 3 = 6$, so the bracket is $8 - 6 = 2$. Then $3 \times 2 = 6$, and $6 + 5 = 11$.

39. (B) Distribute: $7 \times (10 - 3) = 7 \times 10 - 7 \times 3 = 70 - 21 = 49$. "$7 \times 10 - 3$" only multiplies the first term.

40. (D) Bracket first: $2 + 3 = 5$. Then ÷ and × left to right: $60 \div 5 = 12$, then $12 \times 2 = 24$. (Doing × before ÷ to get $60 \div 10 = 6$ is the trap.)

41. (A) With only + and − (same precedence), work left to right: $18 - 5 = 13$, then $13 + 3 = 16$. So the brother's 16 is correct. Reema wrongly treated $5 + 3$ as a bracketed sum to subtract ($18 - 8 = 10$).

42. (C) Factor out 25: $25 \times 4 + 25 \times 6 = 25 \times (4 + 6) = 25 \times 10 = 250$.

43. (B) Values: $2 + 3 \times 4 = 14$, $2 \times 3 \times 4 = 24$, $2 \times 3 + 4 = 10$, $(2 + 3) \times 4 = 20$. The greatest is $2 \times 3 \times 4 = 24$.

44. (D) Innermost: $2 \times 5 = 10$, so the bracket $12 - 10 = 2$. Then $3 \times 2 = 6$, and $100 - 6 = 94$. (Subtracting before multiplying, $97 \times \dots$, or $100 - 36 = \dots$ are sign/order traps.)

45. (A) Minus before the bracket flips both inner signs: $15 - (7 - 2) = 15 - 7 + 2 = 10$. The -2 inside becomes $+2$.

46. (B) Pens cost 8×12 and notebooks cost 5×20; total bill $= 8 \times 12 + 5 \times 20 = 96 + 100$ = ₹196. The other forms mismatch the quantities and prices.

47. (C) $1000 \times 8 - 8 = 8 \times (1000 - 1) = 8 \times 999 = 999 \times 8$ by distributivity. They are equal — no product needed.

48. (D) Bracket first: $6 + 1 = 7$. Terms: $4 \times 5 = 20$ and $2 \times 7 = 14$. So $20 - 14 = 6$.

49. (B) Removing a bracket after a minus flips every inner sign: $80 - (20 - 5) = 80 - 20 + 5 = 65$. ($80 - 20 - 5 = 55$ forgets to flip the -5.)

50. (A) Five days at ₹25 each: 5×25 = ₹125.

51. (C) Innermost bracket: $2 + 1 = 3$. Then $4 \times 3 = 12$, so $3 + 12 = 15$. Finally $2 \times 15 = 30$.

52. (D) $45 \times 99 = 45 \times (100 - 1) = 45 \times 100 - 45 = 4500 - 45 = 4455$. "$45 \times 100 - 1$" subtracts only 1, not 45.

53. (A) Same precedence, left to right: $36 \div 4 = 9$, then $9 \div 3 = 3$. (Division is not associative: $36 \div (4 \div 3) \neq 3$.)

54. (B) $12 - 4 - 2 = 6$ and $12 - (4 + 2) = 12 - 6 = 6$ — equal, since subtracting two numbers equals subtracting their sum. The others differ: $(12 - 4) - 2 = 6$ but $12 - (4 - 2) = 10$; and $12 \div 4 \div 2 = 1.5$ vs $12 \div (4 \div 2) = 6$.

55. (D) Terms: 7, $+(3 \times 4) = +12$, $-(5 \times 2) = -10$. Sum $= 7 + 12 - 10 = 9$.

56. (C) Minus before the bracket flips both inner signs: $64 - (24 + 16) = 64 - 24 - 16 = 24$. ($64 - 24 + 16 = 56$ fails to flip the $+16$.)

57. (A) Bracket first: $4 - 2 = 2$. Terms: $2 \times 5 = 10$ and $3 \times 2 = 6$. So $10 + 10 - 6 = 14$.

58. (B) Pair numbers that make round tens: $8 + 2 = 10$ and $17 + 13 = 30$, giving $10 + 30 = 40$ — easiest mentally. The other pairings (25 & 15, or 21 & 19) are not round tens.

59. (C) First bracket: $3 \times 4 = 12$, so $15 - 12 = 3$. Second: $6 \div 2 = 3$, $+1 = 4$, then $\times 2 = 8$. Total $= 3 + 8 = 11$.

60. (D) $\div$ and $\times$ before $-$ and $+$: $12 \div 4 = 3$ and $2 \times 5 = 10$. Then $36 - 3 + 10 = 43$. (Left-to-right gives 36, the trap in option A.)

Solutions complete: 60 / 60.